University Course Handbook

Generated System

February 17, 2026

Programme Core Mandatory

Kalasalingam Academy of Research and Education

(Deemed to be University)
Under Sec. 3 of UGC Act 1956
Anand Nagar, Krishnankoil (Via),
Virudhunagar (Dt) – 626126, Tamil Nadu

ECE000 SEMICONDUCTOR PHYSICS

ECE000 Semiconductor Physics	L	T	P	X	C
	3	0	2	3	5.0
Pre-requisite: ECE203 Sample course as pre-requistie **Co-requisite:** ECE202 Sample corse as co-requistie	**Category / Type:** PCM / IC-T				

Course Description

This course introduces the fundamental physical principles governing charge transport, energy bands, and carrier dynamics in semiconductor materials. Emphasis is placed on mathematical modeling of electronic behavior using equations such as the drift–diffusion current density $J = qn\mu E + qD\frac{dn}{dx}$, bandgap energy E_g, and Fermi–Dirac statistics $f(E) = \frac{1}{1+e^{(E-E_F)/kT}}$. Analytical treatment of electric fields, carrier concentration profiles, and temperature-dependent conductivity enables students to understand modern electronic and photonic devices at the microscopic level.

Course Objectives

- To evaluate the efficiency (η) of p–n junctions, where power and energy are calculated using $P = I^2 \times R$, while ensuring that 100% of the hash parameters (#), underscores (_), and curly braces ({}) are rendered correctly without breaking Jinja2 templating or LaTeX comments.
- To derive and apply charge transport equations involving drift velocity $v_d = \mu E$ and diffusion current.
- To analyze carrier statistics using Maxwell–Boltzmann and Fermi–Dirac distributions.
- To apply differential equations such as $\frac{d^2V}{dx^2} = -\frac{\rho}{\varepsilon}$ in electrostatic modeling of devices.
- To interpret the role of intrinsic and extrinsic carrier concentrations n_i, n, and p in device operation.

Course Outcomes

CO1: Explain semiconductor energy band diagrams, carrier concentration n, p, and bandgap energy E_g.
CO2: Apply drift and diffusion equations $J_n = qn\mu_n E + qD_n\frac{dn}{dx}$ to analyze current flow.
CO3: Analyze carrier statistics using integrals of the form $\int_{E_c}^{\infty} g(E)f(E)\,dE$.
CO4: Solve electrostatic field problems using Poisson's equation $\nabla^2 V = -\frac{\rho}{\varepsilon}$.
CO5: Evaluate temperature-dependent behavior using $n_i \propto T^{3/2}e^{-E_g/2kT}$.

CO-Blooms Mapping

CO	Course Outcome	Bloom's Level
CO1	Explain semiconductor energy band diagrams, carrier concentration n, p, and bandgap energy E_g.	K2 - AP
CO2	Apply drift and diffusion equations $J_n = qn\mu_n E + qD_n\frac{dn}{dx}$ to analyze current flow.	K3 - CR
CO3	Analyze carrier statistics using integrals of the form $\int_{E_c}^{\infty} g(E)f(E)\,dE$.	K1 - RE
CO4	Solve electrostatic field problems using Poisson's equation $\nabla^2 V = -\frac{\rho}{\varepsilon}$.	K1 - UN
CO5	Evaluate temperature-dependent behavior using $n_i \propto T^{3/2}e^{-E_g/2kT}$.	K2 - AN

Syllabus

Unit	Contents	Hrs.	CO
1	**Semiconductor Fundamentals**	24	CO1, CO3

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Unit	Contents	Hrs.	CO
	Topics: Energy band formation: E–k diagrams, conductors, semiconductors, insulators; Intrinsic and extrinsic semiconductors: intrinsic semiconductors, extrinsic semiconductors, energy bands; Doping, impurity ionization, and Fermi level: doping, impurity ionization, Fermi level; Carrier generation and recombination: generation mechanisms, recombination mechanisms.	9	
	Experiment: PN Junction Characteristics – Measure the I–V characteristics of a PN junction diode under forward and reverse bias to determine cut-in voltage and saturation current. *Experiment: PN Junction Characteristics* – Measure the I–V characteristics of a PN junction diode under forward and reverse bias to determine cut-in voltage and saturation current.	6	
	X-Activity: Semiconductor Simulation – Simulate energy band diagrams using TCAD tools. Compare simulation results with theoretical E–k diagrams.	9	
2	**Carrier Transport Phenomena**	24	CO2, CO3
	Topics: Carrier transport mechanisms: drift, mobility, diffusion; Carrier generation and recombination: generation mechanisms, recombination mechanisms; Resistivity and governing equations: resistivity of doped semiconductors, sheet resistance, Poisson and continuity equations; Intrinsic and extrinsic semiconductors: intrinsic semiconductors, extrinsic semiconductors, energy bands.	9	
	Experiment: PN Junction Characteristics – Measure the I–V characteristics of a PN junction diode under forward and reverse bias to determine cut-in voltage and saturation current.	6	
	X-Activity: Semiconductor Simulation – Simulate energy band diagrams using TCAD tools. Compare simulation results with theoretical E–k diagrams.	9	
3	**Mathematical Representation**	24	CO3
	Topics: Signal representation: $x(t)$, $x[n]$, frequency-domain interpretation; Current density equations: $J = qn\mu E$, drift current expression; Intrinsic and extrinsic semiconductors: intrinsic semiconductors, extrinsic semiconductors, energy bands; Carrier generation and recombination: generation mechanisms, recombination mechanisms.	9	
	Experiment: PN Junction Characteristics – Measure the I–V characteristics of a PN junction diode under forward and reverse bias to determine cut-in voltage and saturation current.	6	
	X-Activity: Semiconductor Simulation – Simulate energy band diagrams using TCAD tools. Compare simulation results with theoretical E–k diagrams.	9	
4	**Mathematical Representation**	24	CO5, CO4
	Topics: Signal representation: $x(t)$, $x[n]$, frequency-domain interpretation; Current density equations: $J = qn\mu E$, drift current expression; Intrinsic and extrinsic semiconductors: intrinsic semiconductors, extrinsic semiconductors, energy bands; Carrier generation and recombination: generation mechanisms, recombination mechanisms.	9	
	Experiment: PN Junction Characteristics – Measure the I–V characteristics of a PN junction diode under forward and reverse bias to determine cut-in voltage and saturation current.	6	

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Unit	Contents	Hrs.	CO
	X-Activity: Semiconductor Simulation – Simulate energy band diagrams using TCAD tools. Compare simulation results with theoretical E–k diagrams.	9	
5	**Mathematical Representation**	24	CO1, CO3
	Topics: Signal representation: $x(t)$, $x[n]$, frequency-domain interpretation; Current density equations: $J = qn\mu E$, drift current expression; Intrinsic and extrinsic semiconductors: intrinsic semiconductors, extrinsic semiconductors, energy bands; Carrier generation and recombination: generation mechanisms, recombination mechanisms; Carrier generation and recombination: generation mechanisms, recombination mechanisms.	9	
	Experiment: PN Junction Characteristics – Measure the I–V characteristics of a PN junction diode under forward and reverse bias to determine cut-in voltage and saturation current.	6	
	X-Activity: Semiconductor Simulation – Simulate energy band diagrams using TCAD tools. Compare simulation results with theoretical E–k diagrams.	9	

Textbook(s)

1. Donald A. Neamen, *"Semiconductor Physics and Devices: Basic Principles"*, 7th Edition, McGraw-Hill Education, 2024.
2. Ben G. Streetman, Sanjay Banerjee, *"Solid State Electronic Devices"*, 4th Edition, Pearson Education, 2015.

Reference(s)

1. `https://nptel.ac.in/courses/106105161`
2. IEEE 802.11-2020, *"Wireless LAN Medium Access Control (MAC) and Physical Layer (PHY) Specifications"*, 2020.
3. M. Stoer and F. Wagner, "A Genetic Algorithm for the Set Covering Problem" *Journal of Heuristics*, Vol. 2, No. 1, 1996.
4. C. Shannon, "A Mathematical Theory of Communication" *Bell System Technical Journal*, Vol. 27, 1948.

Administrative Details

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ECE001 FUNDAMENTALS OF SEMICONDUCTOR PHYSICS FOR VLSI LABORATORY

ECE001 Fundamentals of Semiconductor Physics for VLSI Laboratory	L	T	P	X	C
	0	0	2	0	1.0
	Category / Type: PCM / PC				

Course Description
This course introduces the fundamental concepts of semiconductor physics with emphasis on charge carriers, energy band structure, and material properties. The behavior of electrons and holes, bandgap energy E_g, and temperature-dependent electrical characteristics are analyzed to understand the operation of modern electronic and optoelectronic devices.

Course Objectives
- To analyze the role of doping concentration in semiconductor devices.
- To understand bandgap energy E_g and its impact on electrical conductivity.
- To study bandgap variation in semiconductor materials.
- To analyze α-particle interaction with semiconductor materials.
- To evaluate the effects of temperature on carrier mobility.
- To study the influence of defects in crystalline structures.

Course Outcomes
CO1: Explain the working principles of laboratory tools, sensors, and experimental setups used in practical implementations.
CO2: Apply appropriate measurement and data acquisition techniques to collect and preprocess experimental data accurately.
CO3: Analyze the performance of implemented algorithms or systems using relevant metrics and experimental results.
CO4: Develop and document a complete practical solution by integrating experimentation, analysis, and reporting.

CO-Blooms Mapping

CO	Course Outcome	Bloom's Level
CO1	Explain the working principles of laboratory tools, sensors, and experimental setups used in practical implementations.	K1 - UN
CO2	Apply appropriate measurement and data acquisition techniques to collect and preprocess experimental data accurately.	K2 - AP
CO3	Analyze the performance of implemented algorithms or systems using relevant metrics and experimental results.	K2 - AN
CO4	Develop and document a complete practical solution by integrating experimentation, analysis, and reporting.	K2 - AP

Syllabus

#	Experiment Details	Hrs.	CO
1	**Familiarization with Tools and Setup** Introduction to laboratory instruments, safety procedures, and basic experimental setup.	4	CO1
2	**Measurement and Data Acquisition** Perform measurements using sensors and acquire experimental data for analysis.	6	CO2
3	**Implementation of Core Algorithm** Implement the core algorithm and verify its functionality using test cases.	8	CO2, CO3

#	Experiment Title & Description	Hrs.	CO
4	**Performance Evaluation** Evaluate performance metrics such as accuracy, speed, or efficiency under different conditions.	6	CO3
5	**Case Study and Report Preparation** Apply the developed solution to a real-world case study and prepare a structured report.	6	CO4

Reference(s)

1. Donald A. Neamen, *"Semiconductor Physics and Devices: Basic Principles"*, 2nd Edition, McGraw-Hill Education, 2024.
2. S. M. Sze, *"Physics of Semiconductor Devices"*, 3rd Edition, Wiley, 2018.
3. `www.abc.com/wiki/Semiconductor`

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ECE002 COURSE 2

ECE002 Course 2	L	T	P	X	C
	3	0	2	3	5.0
	Category / Type: PCM / IC-T				

Course Description
Lorem Ipsum has been the industry's standard dummy text ever since the 1500s, when an unknown printer took a galley of type and scrambled it to make a type specimen book. This description serves as placeholder prose to represent a sufficiently detailed course overview.

Course Objectives
- To familiarize students with carrier transport mechanisms and recombination phenomena.
- To provide a foundation for analyzing basic semiconductor devices used in electronic systems.

Course Outcomes
CO1: Explain the concept of energy bands in solids.
CO2: Analyze bandgap energy E_g variation with temperature.
CO3: Explain low-level injection effects in semiconductor materials.
CO4: Analyze α-particle induced defects in silicon.
CO5: Evaluate the impact of device scaling on carrier mobility.

CO-Blooms Mapping

CO	Course Outcome	Bloom's Level
CO1	Explain the concept of energy bands in solids.	K1 - UN
CO2	Analyze bandgap energy E_g variation with temperature.	K2 - AN
CO3	Explain low-level injection effects in semiconductor materials.	K3 - EV
CO4	Analyze α-particle induced defects in silicon.	K3 - CR
CO5	Evaluate the impact of device scaling on carrier mobility.	K1 - RE

Syllabus
Signal Representation Fundamentals
Signals are mathematical functions that convey information about physical phenomena. A continuous-time signal $x(t)$ may be expressed using trigonometric and exponential forms such as $x(t) = A\sin(\omega t + \phi)$, where $\omega = 2\pi f$. Greek symbols like α, β, and λ are commonly used to denote attenuation, phase, and wavelength respectively.
Key Mathematical Relationships
This experiment analyzes angular frequency ω, energy E, and signal power. Greek symbols α, β, and γ are used with equations like $E = mc^2$. $|x(t)|$ is the magnitude of the signal

Feature	Specification	Detail
$\lvert x(t)\rvert$ is the magnitude of the signal	Logic Gate	74LS00
Voltage	5V	Nominal

The table above summarizes key physical quantities.
- Measure frequency using FFT
- Compute energy numerically

This concludes the experiment description.
Properties of Signals
- Linearity enables the use of superposition and scaling.
- Time invariance ensures consistent system response over time.

- Causality implies that the output depends only on present and past inputs.

A clear understanding of these properties is essential for analyzing real-world communication and control systems.

Textbook(s)

1. A. Silberschatz, P. B. Galvin, and G. Gagne, *"Operating System Concepts"*, 10th Edition, Wiley, 2019.
2. B. Forouzan, *"Data Communications and Networking"*, 5th Edition, McGraw Hill, 2017.
3. R. Sedgewick and K. Wayne, *"Algorithms"*, 4th Edition, Addison-Wesley, 2015.
4. J. Kurose and K. Ross, *"Computer Networking: A Top-Down Approach"*, 8th Edition, Pearson, 2021.

Reference(s)

1. `https://nptel.ac.in/courses/106105161`
2. IEEE 802.11-2020, *"Wireless LAN Medium Access Control (MAC) and Physical Layer (PHY) Specifications"*, 2020.
3. M. Stoer and F. Wagner, "A Genetic Algorithm for the Set Covering Problem" *Journal of Heuristics*, Vol. 2, No. 1, 1996.
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